

Lyapunov Stability Theory

Alexander Michailovich Lyapunov

1857-1918

- Russian mathematician and engineer who laid out the foundation of the Stability Theory
- Results published in 1892, Russia
- Translated into French, 1907
- Reprinted by Princeton University, 1947
- American Control Engineering Community Interest, 1960's

Nonlinear Dynamic Systems and Equilibrium Points

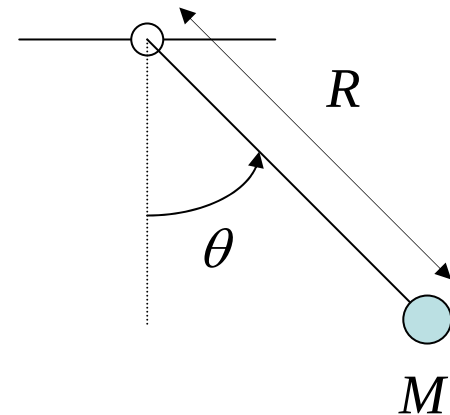
- A nonlinear dynamic system can usually be represented by a set of n differential equations in the form: $\dot{x} = f(x, t)$, with $x \in R^n, t \in R$
 - x is the state of the system
 - t is time
- If f does not depend *explicitly* on time then the system is said to be autonomous: $\dot{x} = f(x)$
- A state x_e is an equilibrium if once $x(t) = x_e$, it remains equal to x_e for all future times: $0 = f(x_e)$

Example: Equilibrium Points of a Pendulum

- System dynamics: $M R^2 \ddot{\theta} + b \dot{\theta} + M g R \sin(\theta) = 0$

- State space representation, $(x_1 = \theta, x_2 = \dot{\theta})$

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{b}{M R^2} x_2 - \frac{g}{R} \sin(x_1)\end{aligned}$$

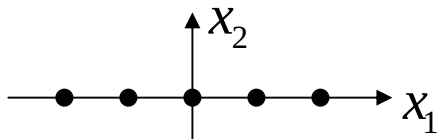


- Equilibrium points:

$$\begin{aligned}0 &= x_2 \\ 0 &= -\frac{b}{M R^2} x_2 - \frac{g}{R} \sin(x_1)\end{aligned}$$

$$x_2 = 0, \quad \sin(x_1) = 0$$

$$x_e = \begin{pmatrix} \pi k \\ 0 \end{pmatrix}, \quad (k = 0, \pm 1, \pm 2, \dots)$$



Example: Linear Time-Invariant (LTI) Systems

- LTI system dynamics: $\dot{x} = Ax$
 - has a single equilibrium point (the origin $\mathbf{0}$) if A is nonsingular
 - has an infinity of equilibrium points in the null-space of A : $Ax_e = 0$
- LTI system trajectories: $x(t) = \exp(A(t-t_0))x(t_0)$
- If A has all its eigenvalues in the left half plane then the system trajectories converge to the origin exponentially fast

State Transformation

- Suppose that x_e is an equilibrium point
- Introduce a new variable: $y = x - x_e$
- Substituting for $x = y + x_e$ into $\dot{x} = f(x)$
- New system dynamics: $\dot{y} = f(y + x_e)$
- New equilibrium: $y = 0$, (since $f(x_e) = 0$)
- Conclusion: study the behavior of the new system in the neighborhood of the origin

Nominal Motion

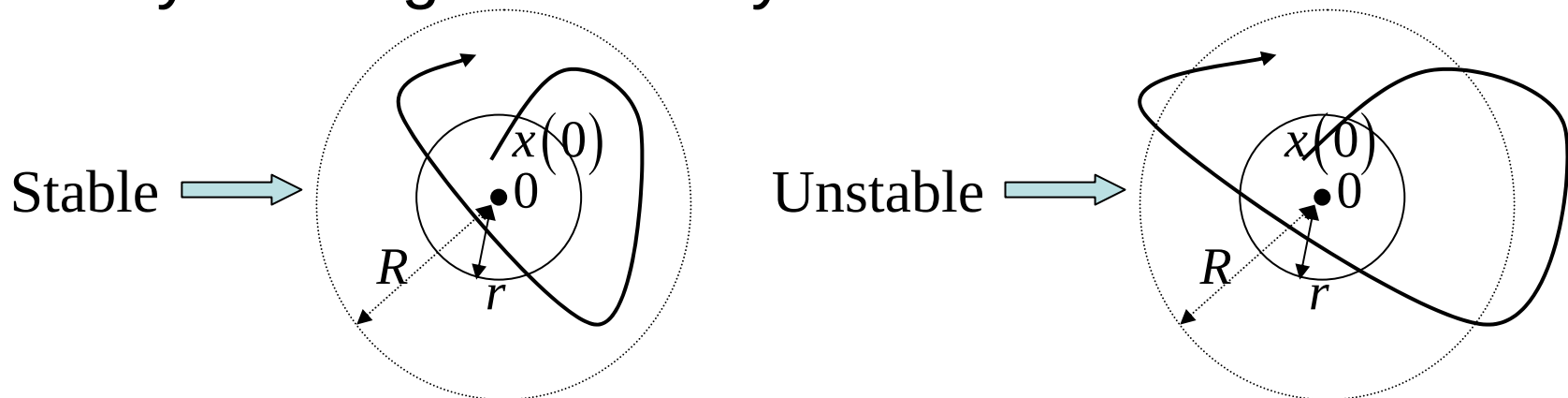
- Let $x^*(t)$ be the solution of $\dot{x} = f(x)$
 - the nominal motion trajectory corresponding to initial conditions $x^*(0) = x_0$
- Perturb the initial condition $x(0) = x_0 + \delta x_0$
- Study the stability of the motion error: $e(t) = x(t) - x^*(t)$
- The error dynamics:
$$\begin{aligned}\dot{e} &= f(x^*(t) + e(t)) - f(x^*(t)) = g(e, t) \\ e(0) &= \delta x_0\end{aligned}$$
- Conclusion: Instead of studying stability of the nominal motion, study stability of the error dynamics w.r.t. the origin

Lyapunov Stability

- **Definition:** The equilibrium state $x = 0$ of autonomous nonlinear dynamic system is said to be stable if:

$$\boxed{\forall R > 0, \exists r > 0, \{ \|x(0)\| < r \} \Rightarrow \{ \forall t \geq 0, \|x(t)\| < R \}}$$

- Lyapunov Stability means that the system trajectory can be kept arbitrary close to the origin by starting sufficiently close to it



Asymptotic Stability

- **Definition:** An equilibrium point $\mathbf{0}$ is asymptotically stable if it is stable and if in addition:
$$\boxed{\exists r > 0, \quad \{\|x(0)\| < r\} \Rightarrow \left\{ \lim_{t \rightarrow \infty} \|x(t)\| = 0 \right\}}$$
- Asymptotic stability means that the equilibrium is stable, and that in addition, states started close to $\mathbf{0}$ actually converge to $\mathbf{0}$ as time t goes to infinity
- An equilibrium point that is stable but not asymptotically stable is called marginally stable

Exponential Stability

- **Definition:** An equilibrium point **0** is exponentially stable if:

$$\boxed{\exists r, \alpha, \lambda > 0, \quad \forall \{ \|x(0)\| < r \wedge t > 0 \}: \quad \|x(t)\| \leq \alpha \|x(0)\| e^{-\lambda t},}$$

- The state vector of an exponentially stable system converges to the origin faster than an exponential function
- Exponential stability implies asymptotic stability

Local and Global Stability

- **Definition:** If asymptotic (exponential) stability holds for any initial states, the equilibrium point is called globally asymptotically (exponentially) stable.
- Linear time-invariant (LTI) systems are either exponentially stable, marginally stable, or unstable. Stability is always global.
- Local stability notion is needed only for nonlinear systems.
- **Warning:** State convergence does not imply stability!

Lyapunov's 1st Method

- Consider autonomous nonlinear dynamic system: $\dot{x} = f(x)$

- Assume that $f(x)$ is continuously differentiable

- Perform linearization:

$$\dot{x} = \underbrace{\left(\frac{\partial f(x)}{\partial x} \right)_{x=0}}_A x + \underbrace{f_{h.o.t.}(x)}_{\text{higher-order terms}} \cong Ax$$

- Theorem**

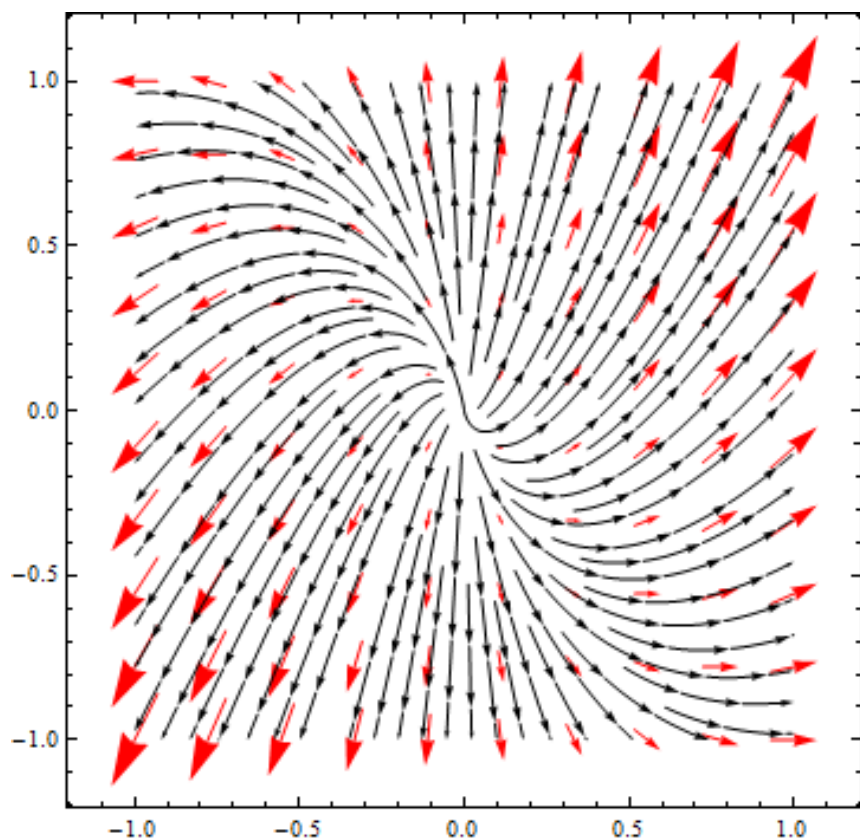
- If A is Hurwitz then the equilibrium is **asymptotically stable**, (locally!)
- If A has at least one eigenvalue in right-half complex plane then the equilibrium is **unstable**
- If A has at least one eigenvalue on the imaginary axis then one **cannot conclude** anything from the linear approximation

Lyapunov's 1st Method - Example

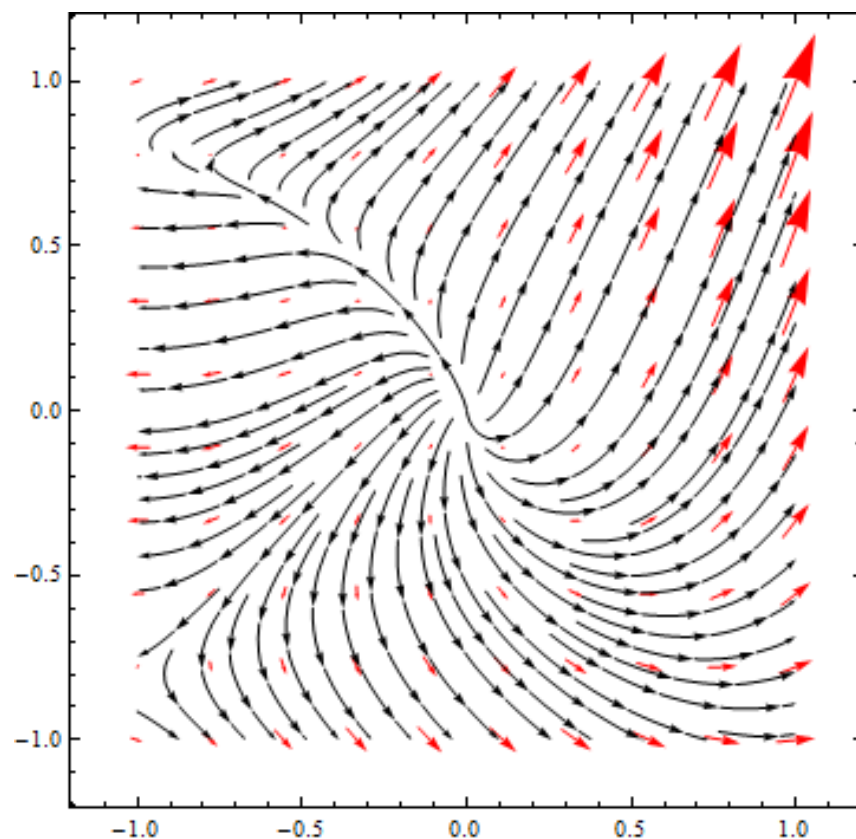
- System Dynamics:
$$\begin{aligned}\dot{x}_1 &= f_1(x_1, x_2) = x_2^2 + x_1 \cos x_2 \\ \dot{x}_2 &= f_2(x_1, x_2) = x_2 + (x_1 + 1)x_1 + x_1 \sin x_2\end{aligned}$$
- Perform linearization around $\mathbf{x} = \mathbf{0}$: $\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x})|_{\mathbf{x}=\mathbf{0}} \mathbf{x}$
$$\mathbf{A}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} \cos x_2 & 2x_2 - x_1 \sin x_2 \\ 2x_1 + 1 + \sin x_2 & 1 + x_1 \cos x_2 \end{bmatrix}, \quad \mathbf{A}(\mathbf{x})|_{\mathbf{x}=\mathbf{0}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$
- Linearized system: $\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \mathbf{x}$
- Eigenvalues: $\lambda_1=1, \lambda_2=1$
- Conclusion: The equilibrium is *unstable*

Lyapunov's 1st Method - Example

Trajectories of the *linearized* system



Trajectories of the *nonlinear* system

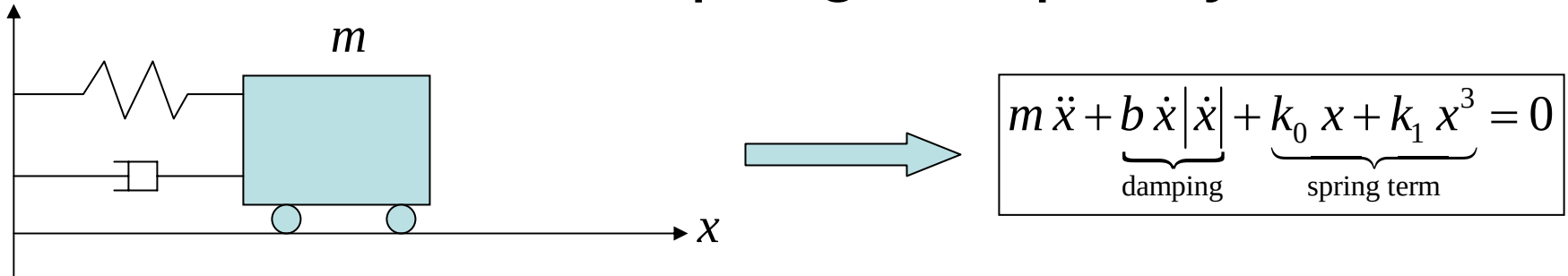


Lyapunov's Direct (2nd) Method

- **Fundamental Physical Observation**
 - If the total *energy* of a mechanical (or electrical) system is continuously dissipated, then the system, *whether linear or nonlinear*, must eventually settle down to an equilibrium point.
- **Main Idea**
 - Analyze stability of an n -dimensional dynamic system by examining the variation of a single *scalar* function, (system energy).

Lyapunov's Direct Method (Motivating Example)

- Nonlinear mass-spring-damper system



- **Question:** If the mass is pulled away and then released, will the resulting motion be stable?
 - Stability definitions are hard to verify
 - Linearization method fails, (linear system is only marginally stable)

Lyapunov's Direct Method (Motivating Example, continued)

- Total mechanical energy

$$V(x) = \underbrace{\frac{1}{2} m \dot{x}^2}_{\text{kinetic}} + \underbrace{\int_0^x (k_0 x + k_1 x^3) dx}_{\text{potential}} = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} k_0 x^2 + \frac{1}{4} k_1 x^4$$

- Total energy rate of change along the system's motion:

$$\dot{V}(x) = m \dot{x} \ddot{x} + (k_0 x + k_1 x^3) \dot{x} = \dot{x} (-b \dot{x} |\dot{x}|) = -b |\dot{x}|^3 \leq 0$$

- Conclusion: Energy of the system is dissipated until the mass settles down: $\dot{x} = 0$

Lyapunov's Direct Method (Overview)

- Method
 - based on generalization of energy concepts
- Procedure
 - generate a scalar “energy-like function (*Lyapunov function*) for the dynamic system, and examine its variation in time, (derivative along the system trajectories)
 - if energy is dissipated (derivative of the Lyapunov function is non-positive) then conclusions about system stability may be drawn

Positive Definite Functions

- **Definition:** A scalar continuous function $V(x)$ is called locally positive definite if

$$V(0) = 0 \wedge \{ \forall x \neq 0 \wedge \|x\| < R \} \Rightarrow V(x) > 0$$

- If $V(0) = 0 \wedge \{ \forall x \neq 0 \} \Rightarrow V(x) > 0$ then $V(x)$ is globally positive definite
- Remarks

– a positive definite function must have a unique minimum

$$\min_{x \in B_R} V(x) = V(x_{\min}) = V_{\min}$$

– if $V_{\min} \neq 0$ or $x_{\min} \neq 0$ then use

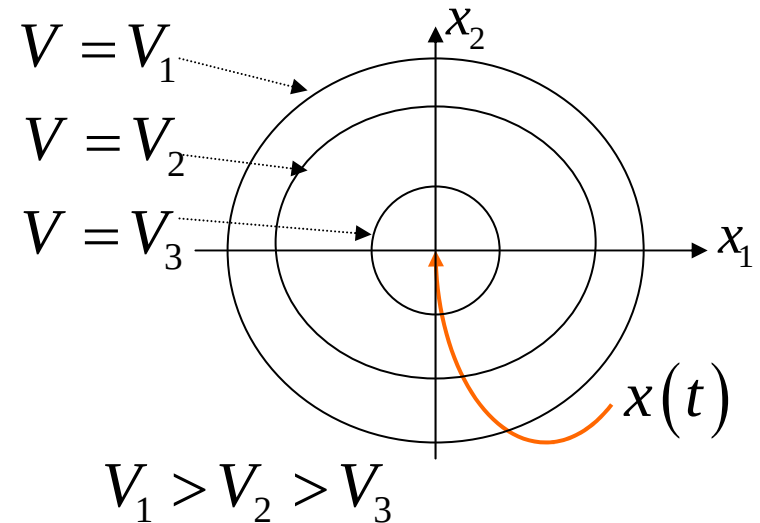
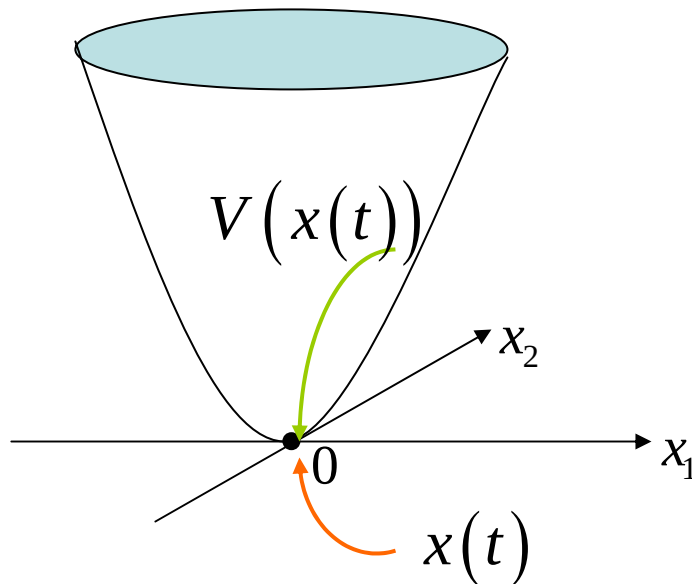
$$W(x) = V(x + x_{\min}) - V_{\min}$$

Lyapunov Functions

- **Definition:** If in a ball B_R the function $V(x)$ is positive definite, has continuous partial derivatives, and if its time derivative along any state trajectory of the system $\dot{x} = f(x)$ is negative semi-definite, i.e., $\dot{V}(x) \leq 0$ then $V(x)$ is said to be a Lyapunov function for the system.
- Time derivative of the Lyapunov function

$$\dot{V}(x) = \nabla V(x) f(x) \leq 0, \quad \nabla V(x) = \left(\frac{\partial V(x)}{\partial x_1} \quad \dots \quad \frac{\partial V(x)}{\partial x_n} \right) \in R^n$$

Lyapunov Function (Geometric Interpretation)



- Lyapunov function is a bowl, (locally)
- $V(x(t))$ always moves down the bowl
- System state moves across contour curves of the bowl towards the origin

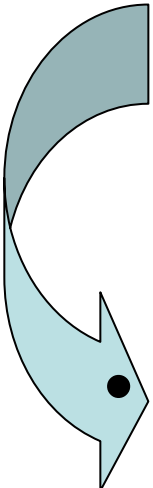
Lyapunov Stability Theorem

- If in a ball B_R there exists a scalar function $V(x)$ with continuous partial derivatives such that $\boxed{\forall x \in B_R : V(x) > 0 \wedge \dot{V}(x) \leq 0}$ then the equilibrium point 0 is stable
 - If the time derivative is locally negative definite $\boxed{\dot{V}(x) < 0}$ then the stability is asymptotic
 - If $V(x)$ is radially unbounded, i.e., $\lim_{\|x\| \rightarrow \infty} V(x) = \infty$, then the origin is globally asymptotically stable
- $V(x)$ is called the Lyapunov function of the system

Example: Local Stability

- Pendulum with viscous damping: $\ddot{\theta} + \dot{\theta} + \sin \theta = 0$
- State vector: $x = (\theta \quad \dot{\theta})^T$
- Lyapunov function candidate: $V(x) = (1 - \cos \theta) + \frac{\dot{\theta}^2}{2}$
 - represents the total energy of the pendulum
 - locally positive definite
 - time-derivative is *negative semi-definite*

$$\dot{V}(x) = \frac{\partial V(x)}{\partial \theta} \dot{\theta} + \frac{\partial V(x)}{\partial \dot{\theta}} \ddot{\theta} = \dot{\theta} \sin \theta + \dot{\theta} \underbrace{\ddot{\theta}}_{-\dot{\theta} - \sin \theta} = -\dot{\theta}^2 \leq 0$$



- **Conclusion:** System is locally stable

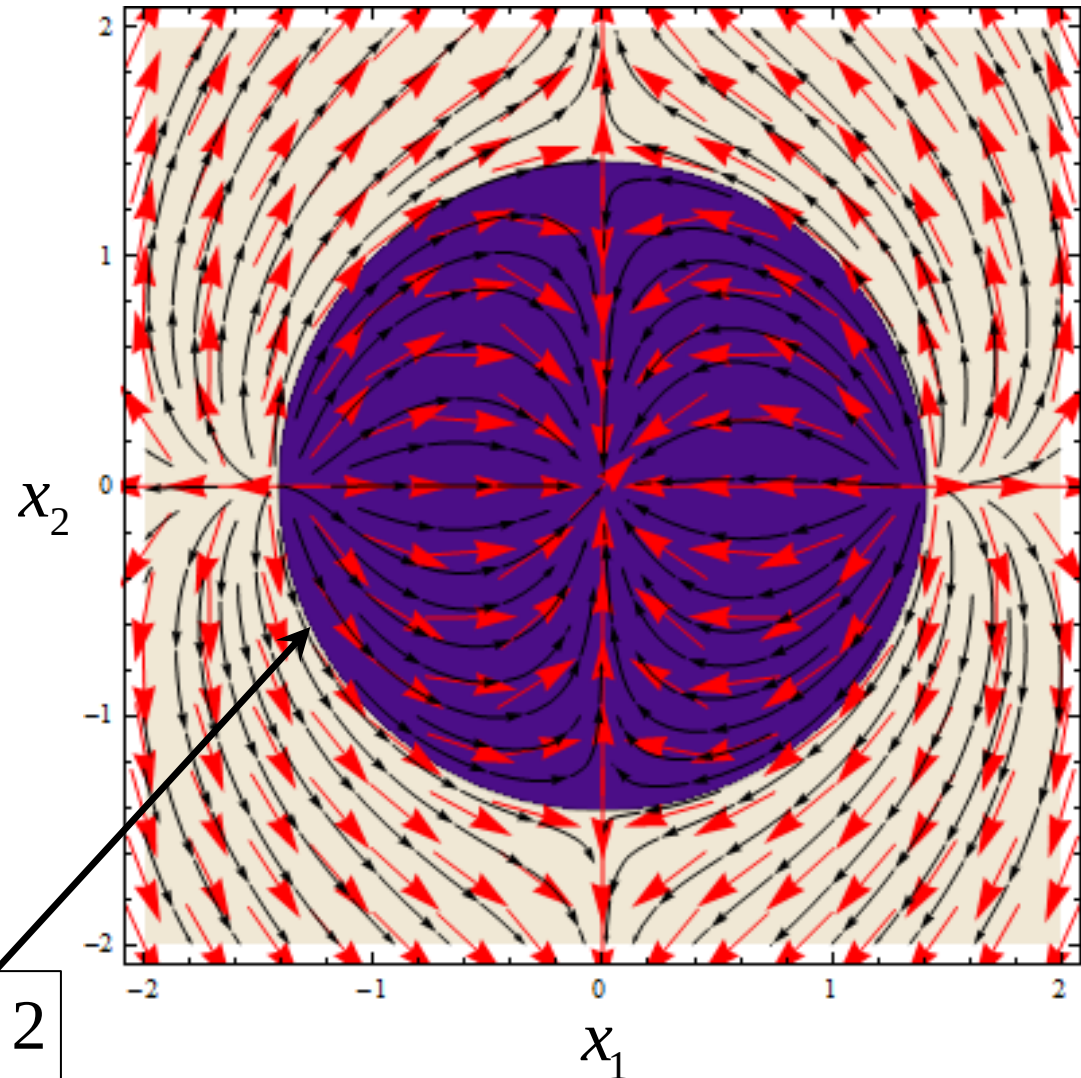
Example: Asymptotic Stability

- System Dynamics:
$$\begin{aligned}\dot{x}_1 &= x_1(x_1^2 + x_2^2 - 2) - 4x_1x_2^2 \\ \dot{x}_2 &= x_2(x_1^2 + x_2^2 - 2) + 4x_1^2x_2\end{aligned}$$
- Lyapunov function candidate: $V(x_1, x_2) = x_1^2 + x_2^2$
 - positive definite
 - time-derivative is *negative definite* in the 2-dimensional ball defined by $x_1^2 + x_2^2 < 2$
$$\dot{V}(x_1, x_2) = 2(x_1^2 + x_2^2)(x_1^2 + x_2^2 - 2) < 0$$
- Conclusion: System is *locally asymptotically stable*

Example: Asymptotic Stability

- System trajectories starting out of the ball $B(0, \sqrt{2})$ are *unstable*
- System trajectories starting in $B(0, \sqrt{2})$ are *asymptotically stable*

$$V(x_1, x_2) = 2$$



Example: Global Asymptotic Stability

- System Dynamics:

$$\begin{aligned}\dot{x}_1 &= x_2 - x_1(x_1^2 + x_2^2) \\ \dot{x}_2 &= -x_1 - x_2(x_1^2 + x_2^2)\end{aligned}$$

- Lyapunov function candidate: $V(x_1, x_2) = x_1^2 + x_2^2$

- Positive definite and radially unbounded
- Time derivative is negative definite

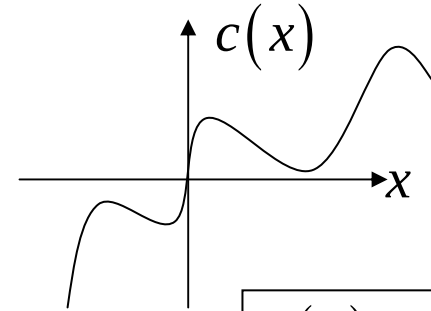
$$\dot{V} = -2(x_1^2 + x_2^2)^2$$

- Conclusion: The origin is a *globally asymptotically stable* equilibrium

Example: Global Asymptotic Stability

- Nonlinear 1st order system

$$\dot{x} = -c(x), \quad \text{where: } x c(x) > 0$$



- Lyapunov function candidate:
 - globally positive definite
 - time-derivative is negative definite

$$V(x) = x^2$$

$$\dot{V}(x) = 2x\dot{x} = -2xc(x) < 0$$

- Conclusion: System is globally asymptotically stable

La Salle's Invariant Set Theorems

- It often happens that the time-derivative of the Lyapunov function is only negative *semi*-definite
- It is still possible to draw conclusions on the *asymptotic* stability
- Invariant Set Theorems (attributed to La Salle) extend the concept of Lyapunov function

Motivating Example: Mass-Spring-Damper System

- System Dynamics: $m\ddot{x} = -b|\dot{x}|\dot{x} - k_0x - k_1x^3$
- Lyapunov function candidate: $V(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}k_0x^2 + \frac{1}{4}k_1x^4$
 - Positive definite
 - Time derivative is *negative semi-definite*

$$\dot{V} = m\dot{x}\ddot{x} + k_0x\dot{x} + k_1x^3\dot{x} = -b|\dot{x}|^3 \leq 0, \quad \dot{V} = 0 \Leftrightarrow \{\dot{x} = 0 \vee x \in \mathbb{R}\}$$

- For $\dot{x} = 0, x \neq 0$ we have $m\ddot{x} = -k_0x - k_1x^3 \neq 0$
 - i.e. $\dot{x} = 0, x \neq 0$ is *not* an equilibrium
 - whereas $\dot{x} = 0, x = 0$ is an equilibrium
- Conclusion: Even if $\dot{V} = 0$ for some non-equilibrium states, the system does not get stuck at them

Invariant Sets & Invariant set theorems

- **Definition:** A set G is an invariant set for a dynamic system if every system trajectory which starts from a point in G remains in G for all future time.
- (*Local Invariant Set Theorem*) Let $V(\mathbf{x})$ be a scalar function with continuous first partial derivatives. Assume that
 - For some $l > 0$, the region Ω_l defined by $V(\mathbf{x}) < l$ is bounded
 - $\dot{V}(\mathbf{x}) \leq 0$ for all $\mathbf{x}$ in Ω_lLet $\mathbf{R}$ be the set of all points within Ω_l where $\dot{V}(\mathbf{x}) = 0$, and $\mathbf{M}$ be the largest invariant set in $\mathbf{R}$. Then, every solution $\mathbf{x}(t)$ originating in Ω_l tends to $\mathbf{M}$ as $t \rightarrow \infty$.
- If $V(\mathbf{x})$ is *radially unbounded*, and $\dot{V}(\mathbf{x}) \leq 0$ over the whole state space, then the result holds *globally*

Invariant Sets & Invariant set theorems

- (*Corollary*) Let $V(\mathbf{x})$ be a scalar function with continuous first partial derivatives. Assume that in a certain neighborhood Ω of the origin
 - $V(\mathbf{x})$ is locally positive definite
 - $\dot{V}(\mathbf{x}) \leq 0$ is negative semi-definite
 - the set $\mathbf{R}$ defined by $\dot{V}(\mathbf{x}) = 0$ contains no trajectories of the system other than the trivial trajectory $\mathbf{x} \equiv \mathbf{0}$

Then, the equilibrium point $\mathbf{0}$ is asymptotically stable.

- Example: Mass-Spring-Damper: $\mathbf{R} = \{ \dot{x} = 0 \vee x \in \mathbb{R} \}$
 - Assume that the largest invariant set $\mathbf{M} \subseteq \mathbf{R}$ contains $\dot{x} = 0, x \neq 0$
Then $\ddot{x} \neq 0$ implies that $\dot{x} \neq 0$, i.e. that the trajectories escape $\mathbf{M}$, a contradiction. Thus $\mathbf{M}$ reduces to $\dot{x} = 0, x = 0$

Example: 2nd Order Nonlinear System

- System dynamics: $\ddot{x} + b(\dot{x}) + c(x) = 0$
 - where $b(x)$ and $c(x)$ are continuous functions verifying the sign conditions:
$$\dot{x} b(\dot{x}) > 0, \text{ for } \dot{x} \neq 0$$
$$x c(x) > 0, \text{ for } x \neq 0$$
- Lyapunov function candidate: $V(x, \dot{x}) = \frac{1}{2} \dot{x}^2 + \int_0^x c(y) dy$
 - positive definite
 - time-derivative is negative *semi*-definite
$$\dot{V} = \dot{x} \ddot{x} + c(x) \dot{x} = -\dot{x} b(\dot{x}) \leq 0$$
 - system energy is dissipated
$$\dot{x} b(\dot{x}) = 0 \Leftrightarrow \dot{x} = 0 \Rightarrow \ddot{x} = -c(x) \Rightarrow x_e = 0$$
 - system cannot get “stuck” at a non-zero equilibrium
- Conclusion: Origin is globally asymptotically stable

Lyapunov Functions for LTI Systems

- LTI system dynamics: $\dot{x} = Ax$
- Lyapunov function candidate: $V(x) = x^T P x$
 - where P is symmetric positive definite matrix
 - function $V(x)$ is positive definite
- Time-derivative of $V(x(t))$ along the system trajectories: $\dot{V}(x) = \dot{x}^T P x + x^T P \dot{x} = x^T \underbrace{(A^T P + P A)}_{-Q} x = -x^T Q x < 0$
 - where Q is symmetric positive definite matrix
 - Lyapunov equation: $A^T P + P A = -Q$
- Stability analysis procedure:
 - choose a symmetric positive definite Q
 - solve the Lyapunov equation for P
 - check whether P is positive definite

Stability of LTI Systems

- **Theorem**

- An LTI system is stable (globally exponentially) if and only if for any symmetric positive definite matrix Q , the unique matrix solution P of the Lyapunov equation is symmetric and positive definite

- **Remark:** In most practical cases Q is chosen to be a diagonal matrix with *positive* diagonal elements

Barbalat's Lemma: Preliminaries

- Invariant set theorems of La Salle provide asymptotic stability analysis tools for autonomous systems with a negative semi-definite time-derivative of a Lyapunov function
- Barbalat's Lemma extends Lyapunov stability analysis to non-autonomous systems, (such as adaptive model reference control)

Barbalat's Lemma

- **Lemma**

- If a differentiable function $f(t)$ has a finite limit as $t \rightarrow \infty$ and if $\dot{f}(t)$ is uniformly continuous, then $\lim_{t \rightarrow \infty} \dot{f}(t) = 0$

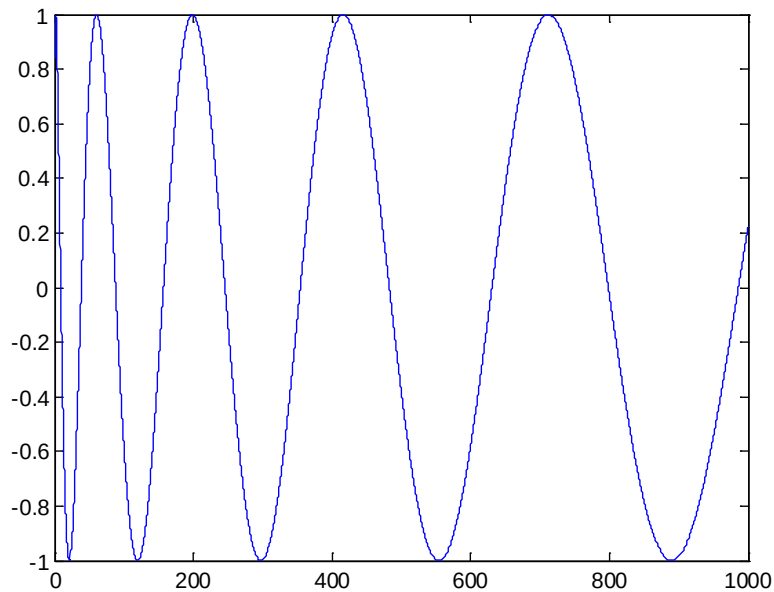
- **Remarks**

- uniform continuity of a function is difficult to verify directly
- simple sufficient condition:
 - if derivative is bounded then function is uniformly continuous
- The fact that derivative goes to zero does not imply that the function has a limit, as t tends to infinity. The converse is also not true, (in general)
- Uniform continuity condition is very important!

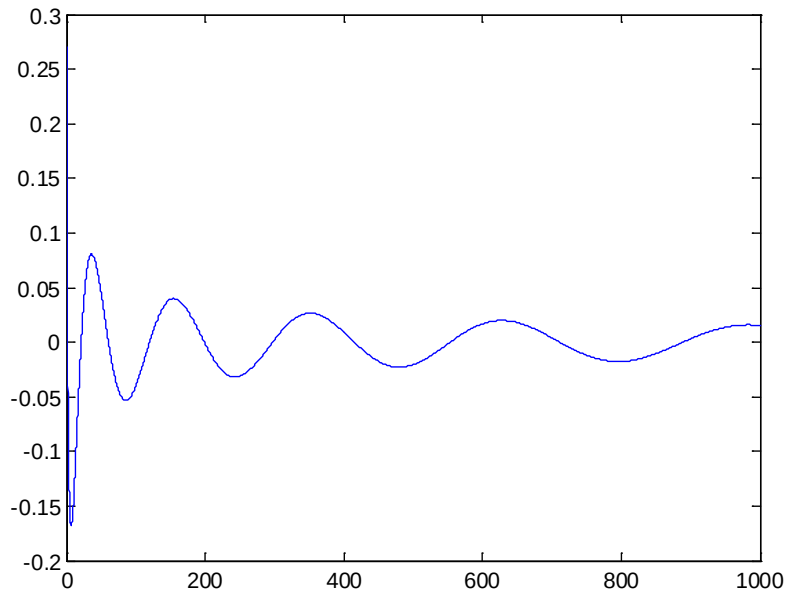
Examples

Fact 1 $\lim_{t \rightarrow \infty} \dot{f}(t) = 0$ *does not* imply that $f(t)$ has a limit as $t \rightarrow \infty$.

$$f(t) = \sin(\sqrt{1+t})$$

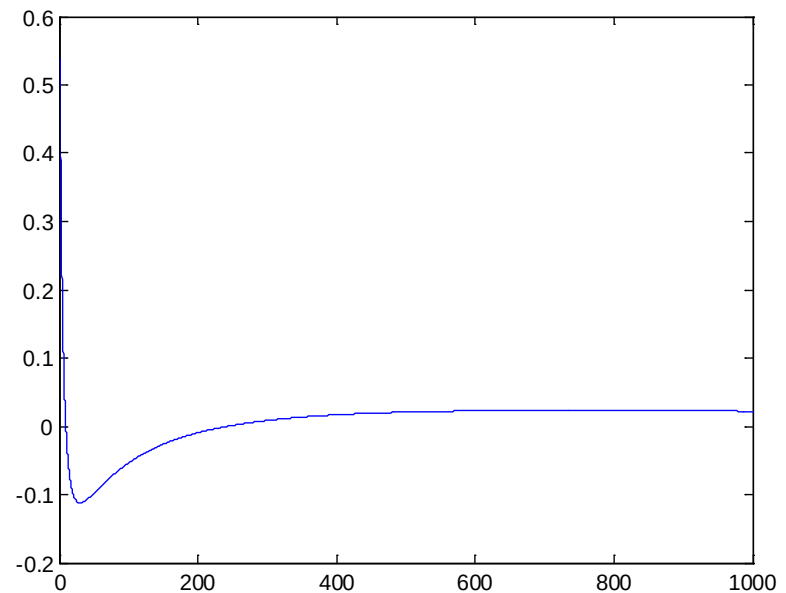
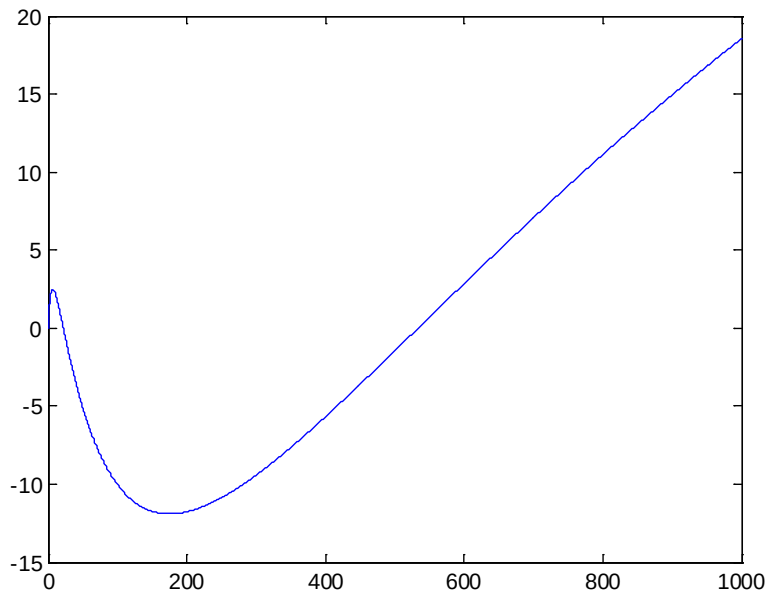


$$\dot{f} = \frac{\cos \sqrt{1+t}}{2\sqrt{1+t}} \rightarrow 0 \text{ as } t \rightarrow \infty$$



Examples

$$f(t) = \sqrt{1+t} \sin(\ln(1+t)) \quad \dot{f}(t) = \frac{\sin(\ln(1+t))}{2\sqrt{1+t}} + \frac{\cos(\ln(1+t))}{\sqrt{1+t}} \rightarrow 0 \text{ as } t \rightarrow \infty$$



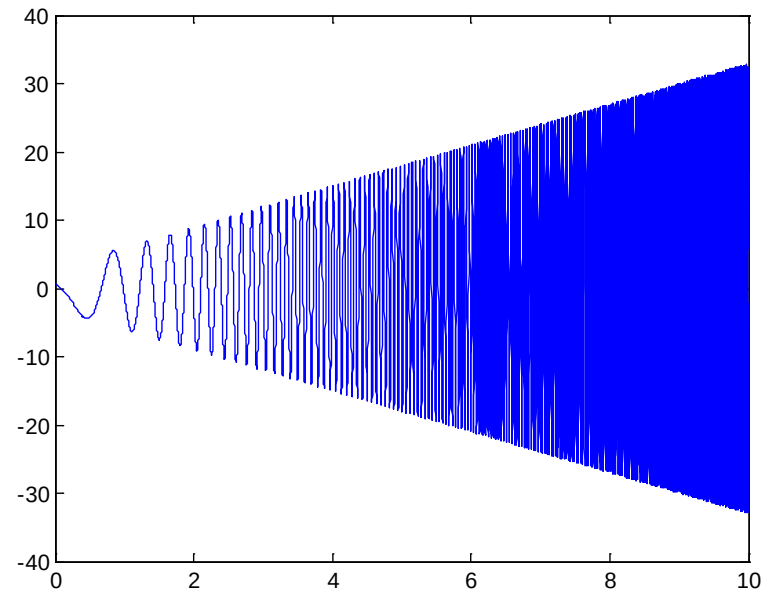
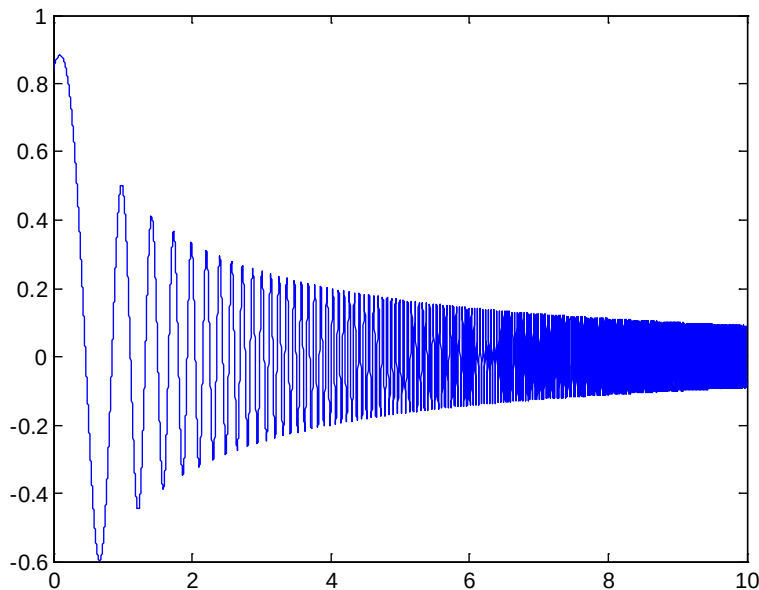
Examples

Fact 2 $\lim_{t \rightarrow \infty} f(t) = c$ for some constant $c \in \mathcal{R}$ *does not* imply that $\dot{f}(t) \rightarrow 0$ as $t \rightarrow \infty$.

$$f(t) = \frac{\sin(1+t)^n}{1+t}$$

$$\dot{f} = -\frac{\sin(1+t)^n}{(1+t)^2} + n(1+t)^{n-2} \cos(1+t)^n$$

$$n=3$$



Lyapunov-Like Lemma

- If a scalar function $V(x,t)$ satisfies the following conditions
 - function is lower bounded
 - its time-derivative along the system trajectories is negative semi-definite and uniformly continuous in time
- Then: $\lim_{t \rightarrow \infty} \dot{V}(x,t) = 0$
- **Question:** Why is this fact so important?
- **Answer:** It provides theoretical foundations for stable adaptive control design

Example: Stable Adaptation

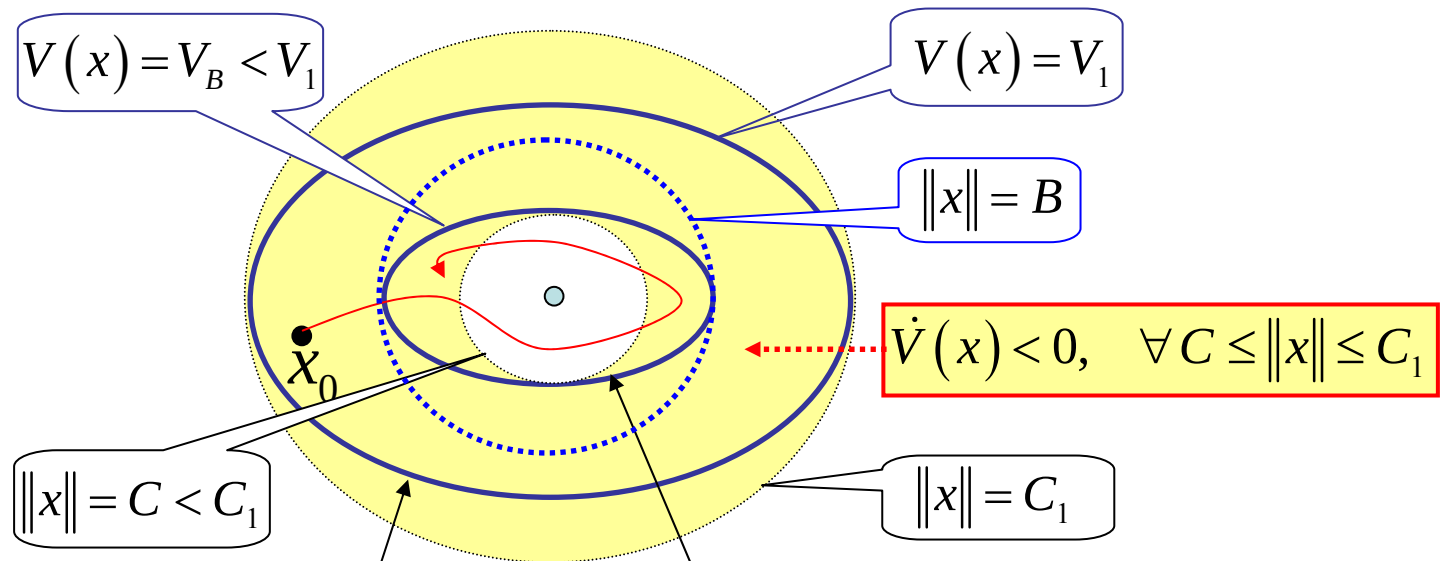
- Closed-loop error dynamics of an adaptive system $\dot{e} = -e + \theta w(t), \dot{\theta} = -e w(t)$
 - where e is the tracking error, θ is the parameter error, and $w(t)$ is a bounded continuous function
- Stability Analysis
 - Consider Lyapunov function candidate: $V(e, \theta) = e^2 + \theta^2$
 - it is positive definite
 - its time-derivative is negative semi-definite
$$\dot{V}(e, \theta) = 2e(-e + \theta w) + 2\theta(-e w) = -2e^2 \leq 0$$
 - consequently, e and θ are bounded
 - since $\ddot{V}(e, \theta) = -4e(-e + \theta w)$ is bounded, $\dot{V}(e, \theta)$ is uniformly continuous
 - hence:
$$\lim_{t \rightarrow \infty} (-2e^2) = \lim_{t \rightarrow \infty} \dot{V}(e, \theta) = 0 \Rightarrow \lim_{t \rightarrow \infty} e(t)$$

Uniform Ultimate Boundedness

- **Definition:** The solutions of $\dot{x} = f(x, t)$ starting at $x(t_0) = x_0$ are Uniformly Ultimately Bounded (**UUB**) with ultimate bound B if:

$$\exists C_0 > 0, T = T(C_0, B) > 0: (\|x(t_0)\| \leq C_0) \Rightarrow (\|x(t)\| \leq B, \quad \forall t \geq t_0 + T)$$

- Lyapunov analysis can be used to show UUB



All trajectories starting in large ellipse enter small ellipse within finite time $T(C_1, B)$

UUB Example : 1st Order System

- The equilibrium point x_e is UUB if there exists a constant C_0 such that for every initial state $x(t_0)$ in an interval $|x(t_0)| \leq C_0$ there exists a bound B and a time $T(B, x(t_0))$ such that $|x(t) - x_e| \leq B$ for all $t \geq t_0 + T$

